
Late Decodability without Behavioral Use: Probing and Patching Recurrent States in Huginn

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Abstract

We test when a known decision-path variable becomes linearly accessible in a fixed-weight depth-recurrent language model. On 500 paired symbolic policy examples, the 3.5B-parameter HUGINN-0125 checkpoint runs for $R \in \{4, 8, 16, 32\}$ loops. Under an identical held-out probe protocol, balanced accuracy is 0.500 in the non-recurrent prelude and 0.473 in the initialized loop-0 state, remains at most 0.581 through loop 8, then reaches 0.689 at loop 9 and 0.919 at loop 16. TF-IDF and frozen-MiniLM monitors of elicited rationales remain statistically compatible with chance at every depth. Yet direct task accuracy is also indistinguishable from chance and does not improve with recurrence. At $R = 32$, paired whole-state patches produce 0/24 donor-directed flips per loop (exact 95% upper limit 0.142), small margin shifts with intervals containing zero, and no separation from unrelated donors; no pair has both baselines correct. Additional recurrence therefore makes the path variable linearly decodable at the probed state, but this checkpoint-task experiment does not show that the model uses that variable to answer or that reasoning migrated from text into latent space.

1 Introduction

Test-time compute can improve language-model performance without changing model weights, but additional compute may also change what an external observer can inspect. Most test-time-scaling systems spend compute in visible token space or search [Snell et al., 2025]; depth-recurrent models instead reuse a latent transformer block before producing the next token [Geiping et al., 2025]. The distinction matters for oversight. Natural-language reasoning can be useful for monitoring behavior [Baker et al., 2025, Korbak et al., 2025], yet generated explanations can omit factors that causally influence an answer [Turpin et al., 2023, Lanham et al., 2023]. If more inference occurs in a recurrent latent state, visible rationales and internal representations may expose different evidence about the computation.

We study this question in HUGINN-0125, an open depth-recurrent model whose `num_steps` argument changes recurrent depth at inference time while keeping weights fixed [Tom Goldstein’s Lab at University of Maryland, 2025, Geiping et al., 2025]. We construct 500 paired symbolic policy examples with a known decision-critical variable: whether the output is determined by a base rule or a multi-hop exception. Crucially, that variable is exactly balanced and decorrelated from the final ALLOW/DENY answer. For each recurrence depth $R \in \{4, 8, 16, 32\}$, we compare a regularized linear probe of recurrent state with TF-IDF and pretrained semantic-encoder monitors of the model’s elicited rationale. We add identical probes at the non-recurrent prelude and initialized loop-0 state, then replace a recipient’s recurrent state with a paired counterfactual donor state at selected loops and measure answer shifts against controls (figure 1).

Contributions and findings. We combine fixed-weight recurrence control, an answer-balanced target, explicit pre-recurrence baselines, two rationale observers, and paired state-patching controls on consumer hardware. Prelude and loop-0 probes are at chance; recurrent-state decodability then

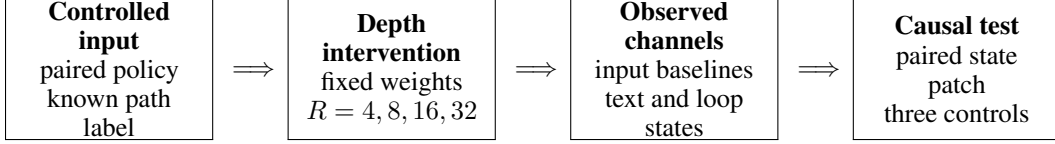


Figure 1: Study design. Recurrence is manipulated without weight updates. Text and latent monitors predict the same ground-truth label; activation patching tests whether a decodable state can also change the valid-answer margin.

changes sharply over loops 9–12 and reaches 0.919 BA at loop 16. Both rationale monitors remain statistically compatible with chance. However, the checkpoint never solves the task above chance, and the narrow patching test is underpowered and nonspecific. The result is a controlled dissociation—late decodability without demonstrated behavioral use—and a negative result about this checkpoint–task pairing, not evidence that reasoning migrated into a hidden chain of thought.

2 Related work

Latent test-time compute. Test-time scaling is commonly implemented through sampling, verification, or iterative refinement [Snell et al., 2025]. Geiping et al. instead train a depth-recurrent transformer in which a shared core receives both an embedded input and a recurrent state [Geiping et al., 2025]. The released HUGINN-0125 checkpoint supports arbitrary `num_steps`, recommends BF16 inference, and exposes the recurrent block in its remote model code [Tom Goldstein’s Lab at University of Maryland, 2025]. Lu et al. analyze Huginn with logit- and coda-lens variants, finding limited evidence for an interpretable latent chain of thought and sensitivity to probe location and decoder [Lu et al., 2025]. We do not decode natural-language tokens from latent states. We measure a task-defined variable with a fixed probe family, and separately test whether paired state replacement affects the answer.

Rationale faithfulness and monitorability. Prompted chain-of-thought explanations can be plausible while failing to disclose biasing causes [Turpin et al., 2023]. Interventions on generated reasoning further show task- and model-dependent causal reliance on the text [Lanham et al., 2023], motivating care in distinguishing faithfulness from output-level consistency [Parcalabescu and Frank, 2024]. At the same time, chain-of-thought monitors can detect reward hacking, while direct optimization against them can incentivize obfuscation [Baker et al., 2025]. A broader multi-organization review argues that this monitoring channel is useful but fragile [Korbak et al., 2025]. Recent work operationalizes monitorability as predicting behavior-relevant properties and separates intervention, process, and outcome-property evaluations [OpenAI, 2025]. Our “rationale” is merely prompt-elicited explanatory text after redacting the requested final-answer field; it is neither a privileged private channel nor assumed to be a faithful report of internal computation.

Probing and causal intervention. Linear decodability establishes predictive accessibility to a probe, not use by the model. Control tasks and restricted probe capacity help contextualize probe scores [Hewitt and Liang, 2019]. Activation patching complements probing by replacing an internal activation from one run with that of another; causal tracing has localized factual associations [Meng et al., 2022], while methodological studies show that effects depend strongly on the intervention and metric [Zhang and Nanda, 2023]. We therefore predefine the state, token, loop, donor construction, score, and three controls. Related work has also combined activation patching with natural-language explanation faithfulness [Yeo et al., 2024]; our focus is the recurrence-depth trajectory and a common text/latent target.

75 3 Study design

76 3.1 Model and recurrence manipulation

77 Let $x = (x_1, \dots, x_n)$ be a prompt, $e = P(x)$ its non-recurrent prelude representation, and h_0
 78 Huginn’s seeded random initial state. The shared recurrent core G_θ and coda C_θ compute

$$h_0 \sim \mathcal{N}_{\text{trunc}}(0, \sigma^2 I), \quad h_k = G_\theta(h_{k-1}, e), \quad k = 1, \dots, R, \quad (1)$$

$$z_R = C_\theta(h_R), \quad p_\theta(v \mid x, R) = \text{softmax}(z_R)_v. \quad (2)$$

79 We vary only R over $\{4, 8, 16, 32\}$. Every condition uses the same checkpoint commit, BF16
 80 weights, batch size one, evaluation mode, and no gradients. The per-example seed resets Python,
 81 NumPy, PyTorch, and MPS before each condition so common recurrence prefixes are coupled. For
 82 every example, independently collected prefixes from smaller and larger R must agree within 10^{-3}
 83 absolute and relative tolerance.

84 3.2 Paired symbolic task

85 Each example describes a base binary policy and a separate multi-hop exception. The base prop-
 86 erty determines an initial ALLOW/DENY decision; if the relation-chain endpoint has the exception
 87 property, the decision is inverted. The target $y^{\text{crit}} \in \{\text{BASE}, \text{EXCEPTION}\}$ records which path de-
 88 termines the final answer. Three surface families – employee access, package routing, and device
 89 authorization – share the abstract program but vary names, property words, rule/fact order, and dis-
 90 tractors.

91 Every pair is assigned wholly to train, validation, or test. A FLIP pair changes one endpoint fact,
 92 changing both the critical label and answer; it is the primary unit for causal patching. A SAME-
 93 ANSWER pair changes base polarity and endpoint status together, changing the critical label while
 94 preserving the final answer. The latter is explicitly a matched contrast, not a minimal one-fact
 95 counterfactual. The generator symbolically re-solves every item, validates pair consistency and
 96 prompt length with the pinned tokenizer, and rejects imbalance. The 500-example default contains
 97 250 pairs, exactly 250 instances per answer and per critical label, with measured answer–critical-
 98 label correlation zero. Split and family counts are reported in appendix B.

99 3.3 Direct scoring and elicited rationales

100 The checkpoint-provided chat template wraps every task prompt using the model card’s recom-
 101 mended system/user/assistant format; the rendered text is encoded without adding a duplicate
 102 special-token layer. The direct prompt maps “A” to ALLOW and “B” to DENY. Tokenizer cali-
 103 bration confirms that the contextual continuations “ A” and “ B” are distinct single tokens. Thus
 104 the valid-answer score for $a \in \{A, B\}$ is the next-token logit $s_R(a) = z_R[a]$; the implementation
 105 falls back to complete teacher-forced continuation log probability if this calibration ever fails.

106 A pilot syntax check found that the pretraining-only checkpoint consistently preferred an answer-
 107 first response. We therefore freeze a prompt that requests FINAL: A or FINAL: B first, followed
 108 by one to three short rationale sentences. Greedy generation uses at most 64 new tokens and stops
 109 after a sentence-final punctuation mark only once at least 12 tokens have been generated; the check-
 110 point end token can also terminate it. Exact-label compliance is reported separately from rationale
 111 availability. Before monitoring, every complete line containing FINAL: is removed wherever the
 112 model placed it; all remaining nonempty generated text is retained. Thus malformed outputs remain
 113 losslessly available in the raw table but cannot leak the parsed final field to the monitor. The monitor
 114 never sees the prompt, label, symbolic rationale, parsed final answer, or activation.

115 3.4 Latent and text monitors

116 At loop k we record the final prompt-token vector $q_{R,k} = h_k[-1] \in \mathbb{R}^{5280}$ after the last sub-block of
 117 the recurrent core. A hook must fire exactly R times and return the model-derived width. Captures
 118 remain on-device until the forward synchronizes, then move to CPU and are stored as float16 arrays
 119 of shape $[N, R, 5280]$. Finite-value, nonzero-state, shape, count, and cross-depth coupling checks
 120 are mandatory. To test whether recurrence creates linear accessibility rather than merely preserving
 121 prompt features, an additional post-protocol pass records $e[-1]$ after the final prelude block and

122 $h_0[-1]$ from the first half of the recurrent adapter input. An independent hook captures the adapter’s
 123 copy of e and must agree exactly within the same tolerance.

124 For each (R, k) , a scaled L2-regularized logistic probe estimates

$$\hat{p}(y^{\text{crit}} = 1 \mid q_{R,k}) = \sigma(w^\top q_{R,k} + b), \quad (\hat{w}, \hat{b}) = \arg \min_{w,b} \sum_{i \in \mathcal{T}} \ell_i + \frac{1}{2C} \|w\|_2^2. \quad (3)$$

125 The regularization $C \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$ is selected only on validation balanced accu-
 126 racy, then refit on train plus validation and evaluated once on test. The prelude and loop-0 base-
 127 lines use this identical protocol. A TF-IDF unigram/bigram logistic model follows the same
 128 split and selection policy for rationale text. A stronger post-protocol monitor mean-pools and
 129 L2-normalizes frozen 384-dimensional ALL-MINILM-L6-v2 sentence embeddings [Reimers and
 130 Gurevych, 2019, Wang et al., 2020], then applies the same validation-selected logistic head. Con-
 131 trols are train-majority and shuffled critical labels for latent states; ground-truth-answer-only, exact
 132 parsed-answer-only when fit-able, rationale-length-only, shuffled labels, prompt-only, and leave-one-
 133 surface-family-out diagnostics for text.

134 Let $L(R)$ and $T(R)$ be held-out BA of the final-loop latent probe and TF-IDF rationale monitor on
 135 the same rationale-available test IDs. We retain their difference only as a descriptive summary,

$$\Delta_{\text{mon}}(R) = L(R) - T(R), \quad (4)$$

136 and do not treat it as a distinct construct: because $T(R)$ is nearly flat, the result is primarily a
 137 latent-decodability trajectory. The originally specified slope of this gap on $\log_2 R$ is reported for
 138 completeness, but the loop-resolved transition and input baselines are the primary evidence; a four-
 139 depth slope cannot establish smoothness.

140 3.5 Causal recurrent-state patching

141 On held-out test pairs, at $R = 32$ and $k \in \{1, 2, 4, 8, 16, 24, 32\}$, let r be a FLIP-pair recipient
 142 and d its counterfactual donor. A forward hook clones the recurrent-block output and performs the
 143 single-token whole-vector intervention

$$h'_k[j] = \begin{cases} h_k^d[-1], & j = -1, \\ h_k^r[j], & j \neq -1, \end{cases} \quad h'_t = G_\theta(h'_{t-1}, e^r), \quad t > k. \quad (5)$$

144 If donor answer D differs from recipient answer O , the primary continuous effect is

$$\Delta_{\text{donor}} = [s_{\text{patch}}(D) - s_{\text{patch}}(O)] - [s_{\text{base}}(D) - s_{\text{base}}(O)]. \quad (6)$$

145 We also report donor-directed answer flips, both unconditionally among correct recipients and
 146 strictly when recipient and donor baselines are correct. Controls patch the recipient state
 147 into itself (sham), use a deterministic unrelated donor matched where possible by family and
 148 prompt length, or use a SAME-ANSWER matched contrast. Any sham deviation beyond $0.001 +$
 149 $0.001 \max(|s_A|, |s_B|, 1)$ aborts the run. For control separation, we additionally compute $\Delta_{\text{cf}} -$
 150 Δ_{unrel} on identical recipient-loop rows where both donors have answers opposite the recipient, aver-
 151 age the two recipient directions within each counterfactual pair, and bootstrap pairs. This conditional
 152 paired contrast is reported with its eligible-pair count; it is not silently extrapolated to unrelated
 153 donors sharing the recipient answer.

154 3.6 Uncertainty, matching, and decision rules

155 All headline accuracies use balanced accuracy. Nonparametric 95% intervals use 2,000 fixed-seed
 156 bootstrap resamples; paired quantities resample common example IDs or pairs together. Flip-rate
 157 intervals are exact binomial intervals; for zero events we also report the rule-of-three upper bound
 158 $3/n$. The originally specified H1 screen used the four-depth monitor-gap trend, but we interpret
 159 it alongside the input baselines and the loop-resolved shape. H2 requires a donor-directed effect
 160 beyond sham and meaningfully distinct from unrelated donors; decodability alone is insufficient.

161 H3 is secondary. For each depth pair, common rationale-available test IDs are matched without
 162 replacement on exact direct-answer correctness and within four rationale tokens. We report coverage,
 163 correctness disagreement, and rationale-length standardized mean difference (SMD). A match is
 164 declared inconclusive if coverage is below 0.60 or $|\text{SMD}| > 0.10$; even a valid match does not
 165 randomize architecture or establish a general causal law.

R	Task acc.	Exact final	Text cov.	TF-IDF BA	MiniLM BA	Latent BA	Δ_{mon}	n
4	0.581	0.040	0.792	0.491	0.543	0.546	0.054	55
8	0.541	0.000	0.928	0.471	0.415	0.567	0.096	67
16	0.500	0.968	1.000	0.432	0.419	0.919	0.486	74
32	0.473	0.796	1.000	0.432	0.486	0.865	0.432	74

Table 1: Held-out summary. n is the rationale-available test count at each depth (55, 67, 74, 74); latent and text scores in a row use those same IDs. Δ_{mon} is the descriptive latent-minus-TF-IDF difference, not an independent metric.

4 Results

Generated empirical summary. The following values are generated from saved held-out predictions and are descriptive unless the protocol’s uncertainty and control criteria are met.

Input baselines. Under the identical latent-probe protocol, the prelude representation has BA 0.500 (95% CI [0.383, 0.617]) and the initialized loop-0 state has BA 0.473 ([0.362, 0.586]).

Descriptive depth statistic. For completeness, the originally specified linear slope of Δ_{mon} on $\log_2 R$ is 0.175 (bootstrap 95% CI [0.110, 0.241]); Spearman ρ is 1.000 (exact two-sided permutation $p = 0.083$); the paired calculation uses 50 test IDs. Its interval resamples examples, not the four depth settings, and does not imply a smooth trend.

Causal patching. At the latest evaluated loop, the pair-level donor-margin shift was 0.018 with 0.000 donor-directed flips among 24 eligible recipients (exact 95% upper limit 0.142; rule-of-three bound 0.125).

Matched analysis. For $R = 16$ versus $R = 32$, match coverage was 0.608 and length SMD was -0.091; status: descriptive matched comparison; not an architectural causal estimate.

4.1 Capability and generation diagnostics

Held-out direct-answer accuracy is 0.581 at $R = 4$ (95% CI [0.473, 0.689]), 0.541 at $R = 8$ ([0.419, 0.662]), 0.500 at $R = 16$ ([0.392, 0.608]), and 0.473 at $R = 32$ ([0.351, 0.581]); every interval contains chance (figure 2). More recurrent compute therefore does not improve this checkpoint’s performance on the task. This is not a secondary caveat: it makes checkpoint–task mismatch a live explanation for the entire dissociation and prevents interpreting the probed variable as successful reasoning.

Exact FINAL: compliance is unstable—0.040, 0, 0.968, and 0.796—while nonempty-rationale coverage after answer-line removal is 0.792, 0.928, 1.000, and 1.000. The corresponding test counts are 55, 67, 74, and 74, as stated in table 1; cross-depth paired analyses use only common IDs. On the 50 test IDs with text at every depth, TF-IDF BA is 0.522, 0.476, 0.489, and 0.478, so the lack of a rising text trajectory survives a fixed-cohort sensitivity check. This does not recover information from missing shallow-depth generations, but it bounds the observed selection effect.

4.2 A late recurrent-state decodability transition

The post-protocol non-recurrent baselines directly test whether the critical label was already linearly available before recurrence (figure 3). With the same splits, feature scaling, validation-only C selection, and train-plus-validation refit as every loop probe, prelude BA is 0.500 (95% CI [0.383, 0.617]) and initialized h_0 BA is 0.473 ([0.362, 0.586]). An independent prelude hook and the recurrent adapter’s copy of the prelude agree exactly for all 500 examples.

In the coupled $R = 32$ trajectory, BA stays at most 0.581 through loop 8, then reaches 0.689 at loop 9 (95% CI [0.585, 0.788]), 0.757 at loop 10, and 0.851 at loop 12. It peaks at 0.946 at loop 18 and ends at 0.865. The terminal-depth scores are 0.568, 0.568, 0.919, and 0.865 on all 74 test examples. Thus recurrence does not create information absent from the prompt—the label is a deterministic function of the input—but it reorganizes that information into a linearly accessible final-token state over loops 9–12.

TF-IDF rationale BA is 0.491, 0.471, 0.432, and 0.432 on each depth’s available test IDs. A stronger frozen-MiniLM embedding monitor yields 0.543 (95% CI [0.413, 0.680]), 0.415 ([0.289, 0.536]), 0.419 ([0.310, 0.523]), and 0.486 ([0.377, 0.603]); every interval includes chance (figure 2). The repeated TF-IDF value 0.432 at $R = 16, 32$ is also compatible with sampling noise rather than systematic inversion because its intervals include 0.5, the encoder does not reproduce a stable below-chance pattern, and shuffled/length controls are noisy.

The descriptive latent-minus-TF-IDF gap is 0.054, 0.096, 0.486, and 0.432 on paired eligible IDs: a step followed by a modest decline, not a smooth increase. For completeness, its originally specified slope on $\log_2 R$ is 0.175 per doubling (example-bootstrap 95% CI [0.110, 0.241]) and Spearman $\rho = 1$ (exact $p = 0.083$). The slope interval conditions on four fixed depths and resamples 50 examples; it does not represent uncertainty over depth settings and is not a headline inferential claim.

4.3 Causal patching is underpowered and nonspecific

The causal results do not establish selective use of the decoded variable (figure 4). At every patched loop, exactly 24 recipient runs prefer their ground-truth answer and are eligible for the counterfactual flip analysis. Counterfactual donor-margin means range from 0.0007 at loop 1 to 0.0176 at loop 32, every 95% interval includes zero, and no eligible recipient flips toward the donor. For 0/24 events, the exact two-sided 95% upper limit is 0.142 and the rule-of-three bound is 0.125, so modest flip rates remain compatible with the data. Moreover, no counterfactual pair has both recipient and donor baselines correct; the stricter flip estimand has $n = 0$.

Self-patching reproduces both answer scores exactly, ruling out an implementation-level sham failure. However, the paired contrast $\Delta_{\text{cf}} - \Delta_{\text{unrel}}$ includes zero at every loop; at loop 32 its mean is 0.0097 (95% CI [-0.0506, 0.0662]). Unrelated donors produce one flip at loops 2, 4, and 16 and three at loop 32 among 13 eligible recipients. This pattern is consistent with whole-vector patching acting mainly as nonspecific disruption rather than transferring the critical feature. The experiment therefore rules out only a large, selective effect of this final-token whole-state intervention; it does not establish that recurrent states are causally unused in general.

4.4 Matched comparisons are null or under-covered

Five of six depth pairs retain only 0.091–0.254 of common examples after requiring equal correctness and a rationale-length difference of at most four tokens, below the prespecified 0.60 coverage threshold. Only $R = 16$ versus $R = 32$ passes both balance gates: 45 of 74 examples are retained (coverage 0.608), length SMD is -0.091 , and correctness disagreement before matching is 0.081. On this matched set, TF-IDF rationale-monitor BA changes by -0.071 (95% CI [-0.270, 0.130]). This valid descriptive comparison is compatible with no change; the remaining comparisons are formally inconclusive.

5 Discussion

The main finding is a late change in linear accessibility, not migration of successful reasoning. The decision-path label is at chance for probes of the prelude and initialized state, remains weak through eight updates, and becomes readily decodable over loops 9–12. Because the label is deterministically present in the prompt, the recurrent core is best described as linearizing or reorganizing input information. The checkpoint’s answer accuracy remains at chance, so the simplest behavioral interpretation is that it does not solve this task and the coda does not use the decoded correlate for the requested A/B mapping.

The intervention results sharpen but do not close that interpretation. Counterfactual patches yield no observed flips, yet 24 eligible recipients imply only a 0.142 exact two-sided upper limit, and the strict both-correct analysis is impossible. Occasional unrelated-donor flips suggest nonspecific disruption by whole-vector replacement. Causal information may instead reside at other token positions, in a subspace not transferred faithfully by whole-vector replacement, or at another interface to the coda. The null rules out the preregistered intervention producing a large selective effect; it does not prove that no recurrent state affects behavior.

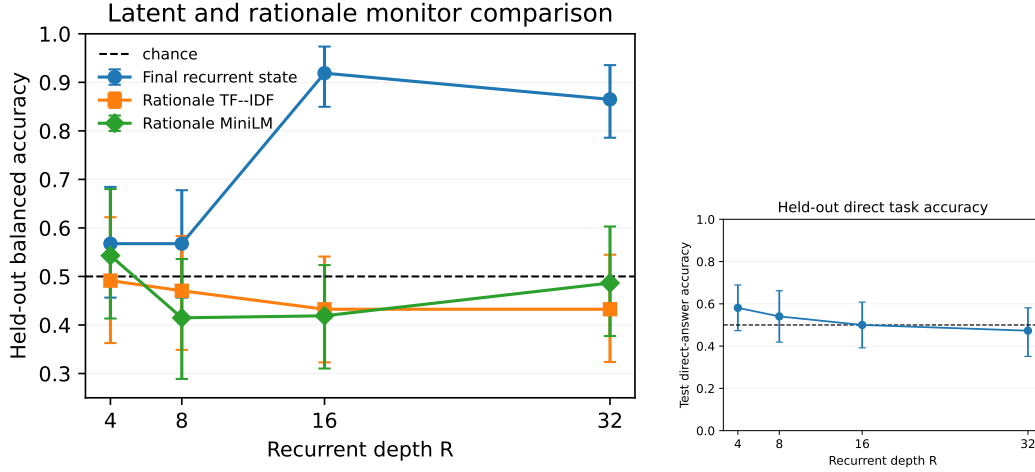


Figure 2: Decodability and capability across recurrence. Left: final-state, TF-IDF rationale, and frozen-MiniLM rationale monitors with 95% bootstrap intervals. Right: held-out direct-answer accuracy. Dashed lines mark chance.

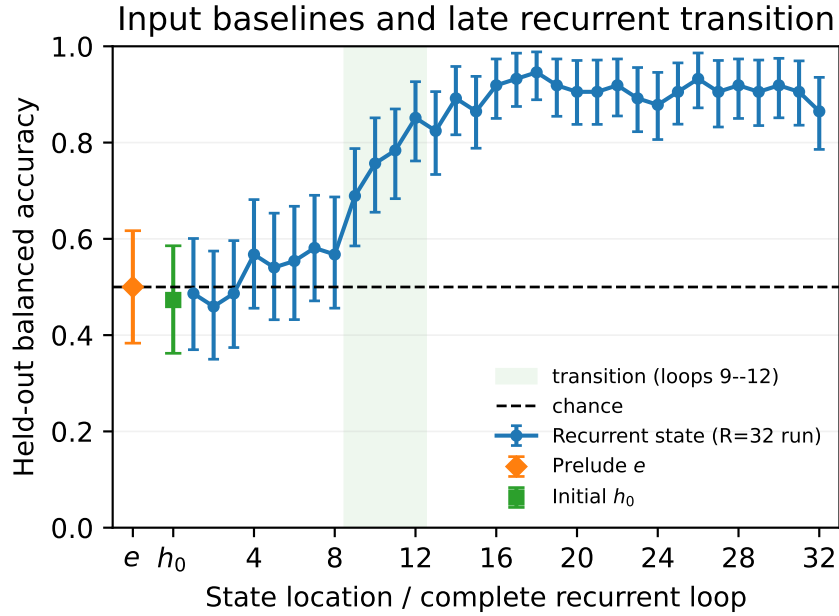


Figure 3: Held-out critical-label probe BA before and during recurrence. Prelude e and seeded h_0 use the identical fit/selection protocol as the loop probes. The longest coupled run is shown because shorter-depth prefixes agree exactly; the shaded region identifies the empirical transition over loops 9–12.

255 The text result is more robust than with TF-IDF alone but remains bounded. A frozen MiniLM
 256 semantic monitor also stays statistically compatible with chance, and a 50-ID common-cohort sen-
 257 sitivity check shows no rising text curve despite changing generation coverage. Nevertheless, nei-
 258 ther observer is capacity-matched to the 5,280-dimensional state probe, and a fine-tuned or larger
 259 language-model monitor could extract different evidence. Below-chance point estimates are not
 260 evidence of obfuscation.

261 The methodological result is therefore a three-way dissociation among input information, linear rep-
 262 resentation, and behavioral use. Follow-up work should repeat the design across checkpoints and

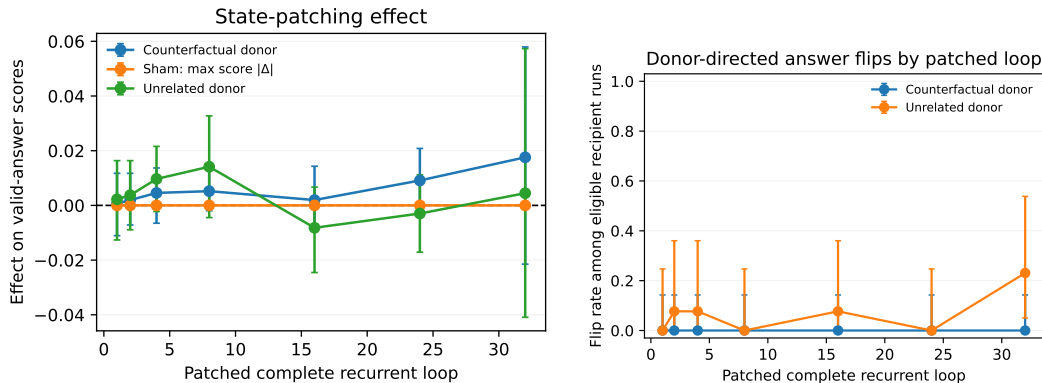


Figure 4: Causal intervention results over patched loop. Donor-margin shifts have pair-bootstrap intervals; flip rates have exact binomial intervals. Counterfactual flips have $n = 24$ eligible correct recipients per loop; no pair has both recipient and donor correct. Unrelated-donor flips indicate nonspecific disruption rather than feature transfer.

seeds, use tasks the model solves reliably, fine-tune stronger text monitors, patch learned probe directions and multiple token positions, and test post-trained recurrent models on naturalistic reasoning. A positive-accuracy replication is required before making a claim about where successful reasoning occurs.

The monitor comparison remains operational rather than capacity-matched. The added MiniLM observer supplies pretrained semantic features and also remains at chance, but a fully fine-tuned or larger language-model judge could behave differently; conversely, a nonlinear latent probe could recover information that is not simply accessible. The reported gap is therefore not an intrinsic quantity of “reasoning.” Causal patches add a different kind of evidence, but a whole 5,280-dimensional state can inject many correlated features. Control separation and temporal localization matter more than a single shift.

6 Limitations and broader impact

The study uses one open 3.5B-parameter pretraining-only checkpoint, one synthetic binary task family, one seed assignment, and one consumer backend. Conclusions may not transfer to post-trained reasoning models, naturalistic tasks, private reasoning channels, other recurrence architectures, or other hardware/dtypes. The fixed dataset is large enough for a pilot but does not average over independently generated datasets or model checkpoints. Test examples share abstract logic with training examples even though surface entities and pairs are separated. Missing or final-only generations can induce selection when rationale availability differs by depth; paired common-ID evaluation reports coverage but cannot recover information from missing text.

Probe accuracy does not identify a mechanism. High dimensionality relative to sample size creates overfitting risk despite linearity, regularization selection, held-out evaluation, answer diagnostics, and shuffled controls. Patching at one token and one hook location is coarse, while unrelated donors are only approximately matched. Only 24 recipient runs per loop are eligible for the primary flip analysis: zero flips has an exact two-sided 95% upper limit of 0.142 (rule-of-three bound 0.125), and no counterfactual pair has both baselines correct. The patching result therefore rules out neither modest flip rates nor causal effects at other tokens/subspaces. BF16/MPS execution is directly verified, but backend-specific kernels may differ from CUDA or CPU. The analysis reports finite-sample uncertainty; it does not correct across every loop-wise exploratory comparison.

The positive use of this work is methodological: monitor developers can test whether changes in inference architecture alter the evidence available to different oversight channels. A negative use is overconfidence – treating latent decoding as mind reading, elicited rationales as faithful private thoughts, or a synthetic result as a safety guarantee. Internal-state monitoring can also create privacy and surveillance concerns if applied to user data or deployed systems. We release no new trained

297 model and use only procedurally generated scenarios with no people, private records, or human
298 subjects.

299 7 Conclusion

300 In HUGINN-0125, a deterministic prompt variable is not linearly decodable from the prelude or
301 initialized state but becomes highly accessible over recurrent loops 9–12. Two rationale observers
302 do not show a corresponding increase. The checkpoint also remains at chance on the task, and
303 an underpowered whole-state intervention shows no selective transfer. The defensible conclusion
304 is therefore narrow: recurrence reorganizes prompt information into a linearly decodable state in
305 this checkpoint–task pairing, without demonstrated behavioral use. Treating that result as hidden
306 chain-of-thought reasoning would outrun the evidence.

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365 A Reproducibility protocol

366 The main run uses HUGINN-0125 revision bb6621b65e90b6a4b9b29ef88dc83866d450470c,
367 PyTorch 2.8.0, Transformers 4.57.3, Python 3.12.8, Apple MPS, and BF16. The model card rec-
368 ommends BF16 and releases the checkpoint under Apache-2.0 [Tom Goldstein’s Lab at University
369 of Maryland, 2025]. Custom Hugging Face code is enabled with `trust_remote_code=True`; pin-
370 ning the full commit SHA bounds this software-supply-chain dependency. No quantization, model
371 fine-tuning, distributed execution, CUDA, or silent device/dtype fallback is used.

372 The complete checkpointed run is:

```
373 PYTHON=./.venv/bin/python bash ./run_all.sh \
374   --config configs/default.yaml \
375   --dataset data/examples.jsonl \
376   --output-root outputs/main \
377   --revision bb6621b65e90b6a4b9b29ef88dc83866d450470c \
378   --device mps --dtype bfloat16 --local-files-only \
379   --allow-main --write-paper-results
```

380 The post-protocol additions run locally from saved artifacts except for one shallow baseline pass:

```
381 ././.venv/bin/python collect_input_baselines.py \
382   --config configs/default.yaml --dataset data/examples.jsonl \
383   --output-dir outputs/main/input_baselines \
384   --device mps --dtype bfloat16 --local-files-only \
385   --set model.revision=bb6621b65e90b6a4b9b29ef88dc83866d450470c
386 ././.venv/bin/python run_reviewer_experiments.py \
387   --config configs/default.yaml \
388   --set model.revision=bb6621b65e90b6a4b9b29ef88dc83866d450470c
```

389 Every inference condition, activation row, and patch intervention checkpoints to Parquet or NPY.
390 Manifests bind outputs to the dataset hash, resolved model revision, resolved configuration hash,

seed policy, answer-token calibration, package versions, and selected backend. Tests cover symbolic generation, token scoring, model loading, recurrent capture count/location, resumability, probe fitting, aggregation, and sham patching.

The audited run used a 12-core Apple M4 Pro (8 performance and 4 efficiency cores) with 24 GB unified memory on macOS 26.5.2. The 7.13 GB model footprint and activations remained on Apple MPS in BF16; saved activation arrays use float16 and occupy 302.12 MiB. The complete output directory is approximately 303 MiB, excluding the roughly 15 GB cached checkpoint.

Completion timestamps give approximately 12.1 hours of wall-clock time for the resumable main pipeline: 6 h 45 min for direct scoring and rationale generation, 1 h 37 min for activation extraction, 2 min 37 s for probe and text-monitor fitting, 3 h 38 min for 1,190 state interventions, and 21 s for aggregation and figures. The post-protocol prelude/loop-0 collection adds approximately five minutes of MPS inference (500 one-step forwards); baseline fitting and cached MiniLM encoding add under 20 seconds on CPU. No model weights are trained or regenerated. The final artifact audit takes approximately 8 s. These are single-machine wall times rather than accelerator-hours and include checkpoint writes; interrupted prompt-format pilots and the invalid $N = 8$ capture smoke test are retained separately but were not metered as part of the main run.

The JSON audit records the full dataset, resolved-configuration, model-revision, and MiniLM-revision identifiers. All common-loop activation prefixes and both copies of the prelude baseline agree exactly, and the maximum sham score deviation is zero; exact identifiers and artifact locations are included in the accompanying artifacts.

Component	Frozen setting
Model	HUGINN-0125, pinned 40-character commit
Architecture	3.5B parameters; 4-block recurrent core; width 5,280
Depths	$R \in \{4, 8, 16, 32\}$
Dataset	500 examples; 70/15/15 pair-grouped split
Generation	greedy; 64-token cap; sentence stop after token 12
Activation	final prompt token; every complete loop; float16 disk
State stochasticity	seeded truncated-normal initialization; test-time noise 0
Probe	scaled logistic; validation-selected C
Text monitor	TF-IDF 1–2 grams plus logistic
Semantic monitor	frozen MiniLM embeddings plus logistic (post-protocol)
Input baselines	final prelude token and initialized h_0 (post-protocol)
Patching	held-out pairs at $R = 32$; loops 1, 2, 4, 8, 16, 24, 32
Uncertainty	2,000 fixed-seed bootstrap resamples; 95% CI

Table 2: Reproducibility settings for the main run and labeled post-protocol checks.

B Dataset construction and example

Pair templates first sample the abstract variables needed to meet the exact joint balance. Surface realization then samples split-specific entities and shuffles rule/fact order. The generator retries a template only when token length or a validation invariant fails; it does not filter on model behavior. The rationale ground truth is used solely to validate the symbolic instance and is never supplied to either trained monitor.

Illustrative abstract instance. A green badge sets the base decision to ALLOW. If the employee’s unit ultimately reports to a restricted division, the exception reverses the decision. “Mira has a green badge; Mira works in Unit K; Unit K reports to Division Z; Division Z is restricted.” The critical label is EXCEPTION and the answer is DENY. Its FLIP partner changes only “restricted” to “open,” producing BASE and ALLOW. A SAME-ANSWER contrast changes both badge polarity and endpoint status, preserving the answer while changing the critical label.

The realized dataset has 250 disjoint counterfactual pairs. Pair-grouped splitting produces 350 train, 76 validation, and 74 test examples; the test set contains 48 FLIP and 26 SAME-ANSWER examples. The three surface families contribute 170 employee-access, 166 package-routing, and 164 device-authorization examples. Both the answer and critical-path labels are exactly balanced (250/250),

427 and their binary Pearson correlation is exactly zero. After applying the checkpoint chat template,
 428 direct prompts contain 185–216 tokens (median 200) and rationale prompts contain 197–228 tokens
 429 (median 212), all below the frozen 256-token ceiling.

Split	Examples	Employee	Package	Device
Train	350	118	116	116
Validation	76	26	26	24
Test	74	26	24	24
Total	500	170	166	164

Table 3: Realized pair-grouped split and surface-family counts.

430 C Correctness gates and instrumentation audit

431 Before the main run, a complete $N = 8$, $R \in \{4, 8\}$ smoke pipeline exercised all stages. The
 432 sham gate exposed a subtle invalid capture implementation: moving a recurrent MPS buffer to
 433 CPU from inside the asynchronous hook produced all-zero saved vectors. No main results were
 434 produced from those artifacts. The collector was changed to clone on-device, synchronize after
 435 the complete forward, then transfer to CPU; an all-zero hard failure and regression test were
 436 added. The corrected real-model capture contained nonzero states, and self-patching at loops 1, 2,
 437 4, and 8 reproduced both valid-answer logits exactly. The invalid smoke directory is retained under
 438 outputs/quarantine/ for auditability.

439 Separate syntax-only prompt pilots established the checkpoint chat template, answer-first response
 440 order, and sentence-boundary stopping rule before valid main tables were completed. Raw-prompt
 441 and pre-stopping partial outputs were quarantined rather than mixed with the reported run. These
 442 operational amendments changed model-interface correctness and runtime, not the labels, splits,
 443 depth conditions, monitor families, causal estimands, or hypothesis rules.

444 The main collector enforces the following gates before downstream use:

- 445 1. exactly R hook invocations and shape $[R, 5280]$ for every example;
- 446 2. all values finite and the capture not identically zero;
- 447 3. common-loop states agree across independently run recurrence conditions;
- 448 4. activation and inference manifests share dataset hash and model commit;
- 449 5. every sham intervention stays within the predefined score tolerance; and
- 450 6. all analysis inputs have unique keys, complete schemas, and compatible conditions.

451 D Additional results and controls

452 **Generation diagnostics.** Table 4 separates exact final-field compliance from the explanation text
 453 actually available to the monitor. “Substantive” requires at least three tokens and one alphabetic
 454 character. Only two generations reach the 64-token ceiling, both in the shallow conditions.

R	Exact final	Text coverage	Substantive	Median tokens	Token-cap rate
4	0.040	0.792	0.726	19	0.002
8	0.000	0.928	0.928	18	0.002
16	0.968	1.000	1.000	23	0.000
32	0.796	1.000	1.000	21	0.000

Table 4: Generation and rationale-availability diagnostics over all 500 examples.

455 **Monitor controls.** At the final loop, the answer-label latent diagnostic has balanced accuracy
 456 0.529, 0.463, 0.521, and 0.546 for $R = 4, 8, 16, 32$; shuffled-critical-label probes have 0.514, 0.514,
 457 0.459, and 0.500. For rationale monitoring, length-only controls range from 0.485 to 0.554 and
 458 shuffled-label controls from 0.500 to 0.554. Exact-answer-only controls are unavailable at $R = 4, 8$

because exact formatting is too sparse and are at chance at $R = 16, 32$. A prompt-only diagnostic reaches 0.554 at $R = 16, 32$, showing that the task text itself contains weak linearly recoverable surface evidence; the generated rationale monitor remains at 0.432. Leave-one-surface-family-out rationale monitors vary from 0.333 to 0.615 with broad intervals and no consistent above-chance pattern.

Post-protocol baselines and semantic monitor. The non-recurrent baseline pass uses the same checkpoint, prompts, seeds, MPS/BF16 backend, final-token position, splits, and logistic selection protocol as the recurrent probes; it adds one shallow forward per example and no weight training. The frozen MiniLM monitor is a post-protocol robustness check rather than part of the original hypothesis rule. Its encoder is pinned to revision c9745ed1d9f207416be6d2e6f8de32d1f16199bf; only its regularized logistic head is fit.

Channel	Location/depth	BA	95% CI	Test n
Latent	Prelude e	0.500	[0.383, 0.617]	74
Latent	Initial h_0	0.473	[0.362, 0.586]	74
MiniLM text	$R = 4$	0.543	[0.413, 0.680]	55
MiniLM text	$R = 8$	0.415	[0.289, 0.536]	67
MiniLM text	$R = 16$	0.419	[0.310, 0.523]	74
MiniLM text	$R = 32$	0.486	[0.377, 0.603]	74

Table 5: Post-protocol input baselines and frozen semantic-encoder rationale monitor.

Complete patching summary. Table 6 reports the primary continuous effect, its paired contrast with unrelated donors on jointly eligible recipients, and donor-directed flips. All self-patch score differences are exactly zero. Counterfactual effects use 24 pairs and have 24 eligible correct recipient runs per loop; the exact two-sided interval for 0/24 flips is $[0, 0.142]$ (rule-of-three upper bound 0.125). No pair has both recipient and donor correct, so the strict flip estimand has $n = 0$. Conditional control contrasts use 21 pairs and 26 recipients.

Loop	Δ_{cf}	95% CI	Median	Flip [95% CI]	$\Delta_{cf} - \Delta_{unrel}$	95% CI	Sham max
1	0.0007	$[-0.0111, 0.0117]$	0.0000	0 [0, .142]	-0.0060	$[-0.0208, 0.0082]$	0
2	0.0020	$[-0.0072, 0.0117]$	0.0000	0 [0, .142]	-0.0007	$[-0.0126, 0.0119]$	0
4	0.0046	$[-0.0065, 0.0137]$	0.0000	0 [0, .142]	-0.0074	$[-0.0216, 0.0067]$	0
8	0.0052	$[-0.0020, 0.0130]$	0.0000	0 [0, .142]	-0.0141	$[-0.0283, 0.0015]$	0
16	0.0020	$[-0.0091, 0.0143]$	0.0000	0 [0, .142]	0.0126	$[-0.0037, 0.0313]$	0
24	0.0091	$[-0.0020, 0.0208]$	0.0156	0 [0, .142]	0.0134	$[-0.0030, 0.0290]$	0
32	0.0176	$[-0.0215, 0.0579]$	-0.0156	0 [0, .142]	0.0097	$[-0.0506, 0.0662]$	0

Table 6: Held-out $R = 32$ state-patching results. Effects are changes in the donor-versus-original answer margin; flip intervals are exact binomial.

Matched-depth analysis. The complete matching audit appears in table 7. “Inconclusive” denotes coverage below 0.60; the final row passes the coverage and $|\text{SMD}| \leq 0.10$ gates but its monitor difference includes zero.

Depths	Common	Matched	Coverage	Length SMD	$\Delta_{\text{text BA}}$	Protocol status
4-8	50	12	0.240	-0.222	-0.438	Inconclusive
4-16	55	6	0.109	0.049	-0.125	Inconclusive
4-32	55	5	0.091	0.420	-0.417	Inconclusive
8-16	67	16	0.239	0.186	0.188	Inconclusive
8-32	67	17	0.254	-0.098	0.007	Inconclusive
16-32	74	45	0.608	-0.091	-0.071	Descriptive; CI crosses zero

Table 7: Same-ID matching on equal direct correctness and at most four rationale tokens of difference.

Artifact audit. The fail-closed audit verifies four complete inference conditions, recurrent activation shapes $[500, R, 5280]$, finite nonzero states, exact common-prefix coupling, all 1,190 expected intervention rows, and a maximum sham score difference of zero. Post-protocol gates additionally require complete $[500, 5280]$ prelude and loop-0 arrays, exact agreement between two prelude capture paths, two successful input-baseline probes, and four successful MiniLM conditions. Hashes

484 and locations are recorded in `outputs/main/tables/artifact_audit.json` and the stage man-
 485 ifests.

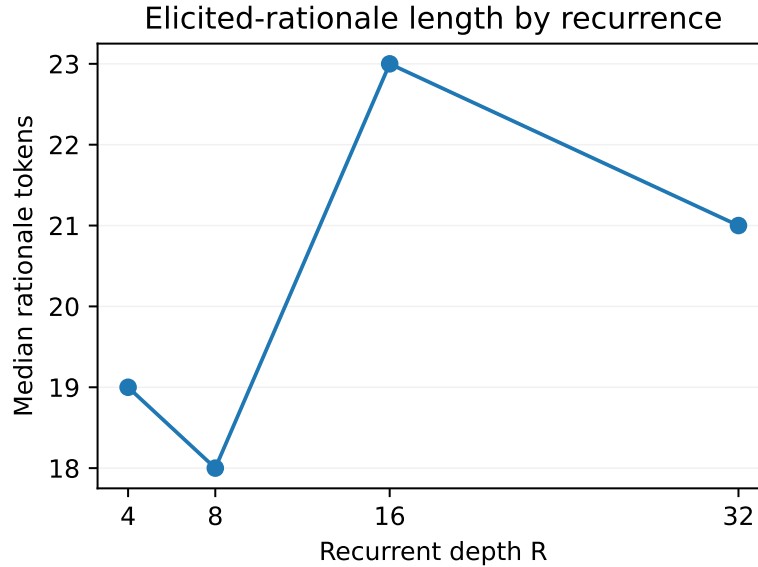


Figure 5: Median nonempty rationale-text length by recurrence depth.

486 E Artifact map

487 The project artifact tree keeps raw observations separate from aggregates. Inference rows are under
 488 `outputs/main/inference/`; dense recurrent arrays under `outputs/main/activations/`; post-
 489 protocol prelude and loop-0 arrays under `outputs/main/input_baselines/`; held-out monitor
 490 predictions under `outputs/main/probes/`; raw interventions under `outputs/main/patching/`;
 491 and CSV summaries plus figures under `outputs/main/tables/` and `outputs/main/figures/`.
 492 Generated manuscript fragments are under `paper/generated/`. The headline table is emitted di-
 493 rectly by the analysis, and every narrative number is cross-checked against a saved table rather than
 494 inferred from a plot.

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